Recovery without a declining opponent: a two-state model motivated by the uncanny valley

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Abstract

An improvement in overall affinity need not mean that an unsettling response has declined: a positive contribution may instead come to outweigh a growing opponent. This note examines that possibility in a bounded two-state model motivated by the uncanny valley. It distinguishes Mori's appearance continuum from the evolution of a response during a fixed exposure. With separate positive rates for acquisition, loss, opponent recruitment and removal, both states increase from a neutral start. Their ratio is strictly increasing when $ab \leq (a+d)^2$ and has one maximum when $ab > (a+d)^2$. In the latter regime, a defined interval of observation weights produces an initial rise, a negative excursion and a return above baseline. A worked example and a complete proof describe how this reversal of relative dominance occurs. A comparison with two real linear decay modes establishes a limited response distinction without selecting a unique mechanism. The measurement analysis also separates parameter uniqueness from prediction: exact opponent and mixed-response traces determine the reduced dynamics despite an unresolved scale of the positive component. The state architecture and general mathematical methods have established precedents, which are identified explicitly. The result is a conditional account of what this model permits, with synthetic illustrations and no participant experiments or empirical parameter fits. It does not establish that the latent components describe human familiarity or eeriness.

Keywords: uncanny valley; opponent processes; dynamical systems; response recovery; structural identifiability; phenomenological modeling

1 The question behind a recovered response

Mori's proposed uncanny valley describes a response across human likeness: affinity rises, falls in an unsettling region, and rises again toward a convincing human appearance [1]. The final rise leaves an interpretive question. An unsettling contribution could weaken, a favorable contribution could grow enough to outweigh it, or both could change. An overall affinity curve does not separate those possibilities.

An earlier article represented affinity as the difference between familiarity and eeriness and used stimulus-dependent rates to produce an appearance-based valley [2]. The purpose of the present note is to examine what that interaction actually requires. In particular, can a recovered response arise from fixed process rules while its negative component continues to grow? A useful answer must give conditions that exclude the response as well as an example that produces it.

We investigate one continuous exposure to a fixed stimulus. This time variable differs from Mori's appearance coordinate: the stimuli are held fixed while two proposed responses develop. Accordingly, the calculation is a temporal extension inspired by the original model, rather than a derivation of Mori's curve from time dynamics. Throughout the main analysis, a *recovered negative excursion* means that the rating initially rises above its neutral baseline, becomes negative relative to it, and later returns above it. An upward movement that remains below baseline is a weaker observation.

The model gives a favorable process an immediate response and lets it recruit a delayed opponent. Sequential recruitment has a precedent in opponent-process theory [3]; the familiarity and eeriness labels are a proposed application of that logic. The two states are latent mathematical quantities. Their interpretation requires more than naming two questionnaire scores, especially because the positive state is assumed both to increase affinity and to drive the opponent.

The analysis follows that proposed interaction through its response. We first distinguish the freedom to draw a static valley from the constraints on a fixed-input trajectory. We then derive the condition under which the opponent gains and loses relative dominance, compare the recovered crossing with a specified linear alternative, and examine what additional observations identify. The fixed-input state equations are contained in Sontag's bilinear feedforward family [4]; the ratio argument uses the same established monotonicity structure as Pinelis's rules [5]. The identification calculation is a model-specific use of the feature-independence criterion of Stanhope, Rubin and Swigon [6]. The note supplies the calculations for this response and clarifies their interpretation, without claiming a new general architecture or mathematical method.

2 From an affinity curve to competing responses

2.1 What the two components represent

A favorable contribution and an unsettling contribution can coexist in the proposed description. Let $H, E \in [0, 1]$ denote their dimensionless magnitudes, retaining the names familiarity and eeriness from the earlier article. The first component contributes positively to affinity; the second opposes it. Their coexistence allows a low rating to arise from little positive familiarity, substantial eeriness, or a combination of both. The model is intended to distinguish these possibilities rather than identify either state with the name of a questionnaire.

At a fixed stimulus, take

$$\dot{H} = a(1 - H) - dH = a - (a + d)H, \tag{1}$$

$$\dot{E} = bH(1 - E) - cE, \tag{2}$$

where $a, d, b, c > 0$ have units of inverse time. Familiarity acquisition acts on the remaining capacity $1 - H$, while loss is proportional to the current familiarity. The opponent is recruited in proportion to H, but only through its remaining capacity $1 - E$; removal acts at rate cE. Thus a larger positive component supplies more input to the opponent, while a larger opponent has less capacity for further recruitment. The evolving balance between recruitment, saturation and removal will determine whether the opponent retains its dominance.

This coupling is a delayed-opposition hypothesis. It represents a positively contributing process that recruits a counteracting response, as in the sequential logic of opponent-process accounts. It does not assert that familiar objects generally become more eerie. Nor is familiarity physically transferred into eeriness: the two states are bounded response magnitudes, not conserved compartments. For that reason familiarity loss dH and eeriness generation $bH(1 - E)$ have independent coefficients. The earlier restriction $d = b$ equates two process scales whose proposed roles differ; below we show which response it excludes.

The rating is

$$Y = o + pH - qE, \qquad p, q > 0, \quad o \in \mathbb{R}. \tag{3}$$

The weights p and q convert the two components into rating units, and o locates the scale's origin. Equal-weight affinity is $F = H - E$. A higher rating can result from a larger positive contribution, a smaller negative contribution, or changes in both. The difference $pH - qE$ records their net effect; it does not determine the direction of either component separately. A reported zero is equally ambiguous, since opposed nonzero contributions can cancel. Both the weights and the initial components will therefore enter the interpretation of a response.

The bounds are preserved by the equations. At $H = 0, 1$, the first derivative is respectively $a, -d$; at $E = 0, 1$, the second is respectively $bH, -c$. The vector field points into the square or along its boundary, giving unique global forward solutions that require no numerical clipping. For later use, write

$$A = a + d, \qquad h = \frac{a}{A}, \qquad B = bh, \qquad K = B + c. \tag{4}$$

Here A is the relaxation rate of familiarity and h its eventual level. Once familiarity has reached that level, B is the opponent's recruitment scale and K its total relaxation coefficient. The equilibrium is $H_* = h$, $E_* = B/K$, and every initial state in the square approaches it. Appendix A gives the convergence argument.

2.2 What appearance and exposure each change

The original uncanny-valley question concerns a human-likeness coordinate R. Within the model, stimulus differences can be represented by functions $a(R), d(R), b(R), c(R)$. If each stimulus is observed after its response has settled, the equal-weight score is negative precisely when

$$b(R)\big[1 - h(R)\big] > c(R). \tag{5}$$

This criterion describes a balance: the opponent must be recruited strongly enough, relative to its removal and the positive component's level, to dominate the net response. A negative region between positive regions requires this balance to reverse across the stimulus continuum. For a weighted rating, the corresponding offset-subtracted condition is $Y_* - o < 0$ exactly when $qb > pK$. Neither sign change is a loss of equilibrium stability.

The sign criterion describes a balance, but a prediction across appearance still requires justified rate functions. If those functions remain unrestricted, they can reproduce any smooth equal-weight equilibrium score between -1 and 1, as Appendix D shows. Drawing a valley in this way therefore does not establish the mechanism that produced it.

To obtain a more constrained question, we now hold one stimulus and all four rates fixed and follow the states in elapsed time t. Neither a scan along the appearance coordinate nor a sequence of encounters is automatically such a trajectory. A scan would also require its speed and initial state; repeated encounters require exposure durations and a rule for carryover. The fixed-exposure calculation isolates changes produced by the state interaction itself.

3 How a rating recovers while its opponent grows

3.1 Following the response through its three phases

Begin with $H(0) = E(0) = 0$, a neutral latent initial state. This is a stated condition on the components, stronger than a reported neutral rating. Acquisition acts immediately on H, whereas recruitment of E must wait for H to become positive. Near the beginning,

$$H(t) = at + O(t^2), \qquad E(t) = \tfrac{1}{2}abt^2 + O(t^3). \tag{6}$$

The positive contribution therefore has an initial lead. Its subsequent course is

$$H(t) = h(1 - e^{-At}), \tag{7}$$

so it keeps increasing while its rate of growth diminishes toward its plateau.

As familiarity develops, the input to its opponent also grows. To see what constrains that growth, rewrite the second equation as

$$\dot{E} = (bH + c)\left(\frac{bH}{bH + c} - E\right). \tag{8}$$

The opponent moves toward the instantaneous target $bH/(bH + c)$. That target rises with H but flattens as recruitment saturates, and the adjustment coefficient $bH + c$ also increases with H. Starting from zero, E stays below this moving target and rises strictly throughout the exposure; the short proof is in Appendix A. Thus an eventual recovery of affinity cannot be attributed to declining E in this neutral-start case. It must come from a reversal in the balance between two increasing contributions.

A constructed example makes that possibility visible. Set

$$a = d = \tfrac{1}{2}, \quad b = 8, \quad c = \tfrac{1}{10}, \quad p = 2, \quad q = 1, \quad o = 0. \tag{9}$$

Initially the rating increases with slope $f'(0) = 1$, where $f(t) = Y(t) - Y(0)$. The recruited opponent then overtakes the positive contribution. At $t = 1$, the calculated contributions are $2H \approx 0.632$ and $E \approx 0.751$, giving $f \approx -0.118$. Later, $2H$ approaches 1 while E approaches $40/41 \approx 0.976$. The score consequently returns above baseline and tends to $1/41$. Eeriness has grown throughout; the positive contribution eventually outweighs it.

Figure 1 separates the component history from the appearance of the net rating. The initial favorable response, unsettling interval and subsequent recovery arise under one fixed set of rates. Within the model, the recovery means that the balance of contributions has changed. If an observed affinity record had this form, its upturn alone would not establish that the unsettling response had weakened. The next step is to determine when this explanation is available, rather than infer a general possibility from one selected example.

3.2 When the opponent can gain and lose dominance

A baseline crossing occurs when the weighted contributions are equal. For $t > 0$, define $z = E/H$ and $z_\infty = b/K$. Then

$$f(t) = Y(t) - Y(0) = qH(t)\left(\frac{p}{q} - z(t)\right). \tag{10}$$

The rating is positive relative to its baseline when $z < p/q$, and negative when $z > p/q$. A recovered negative excursion therefore requires z to rise above the level p/q and later return below it. This is why the ratio, rather than either component alone, determines the response question.

Proposition 1 (Ratio shape and the rate boundary)**.** *Assume constant $a, d, b, c > 0$ and $H(0) = E(0) = 0$. If $ab \le (a + d)^2$, then $z = E/H$ increases strictly from zero to $z_\infty = b/K$. If $ab > (a + d)^2$, then z has exactly one stationary point, a nondegenerate maximum $M > z_\infty$. It increases before that point, decreases afterward, and approaches z_∞ from above.*

The two cases have different behavioral interpretations within the model. If $ab \le A^2$, the opponent keeps gaining relative to familiarity until their ratio settles. Once it has pushed the rating below baseline, it cannot relinquish enough relative dominance to bring it back. If $ab > A^2$, the opponent first gains and then loses ground relative to the positive component,

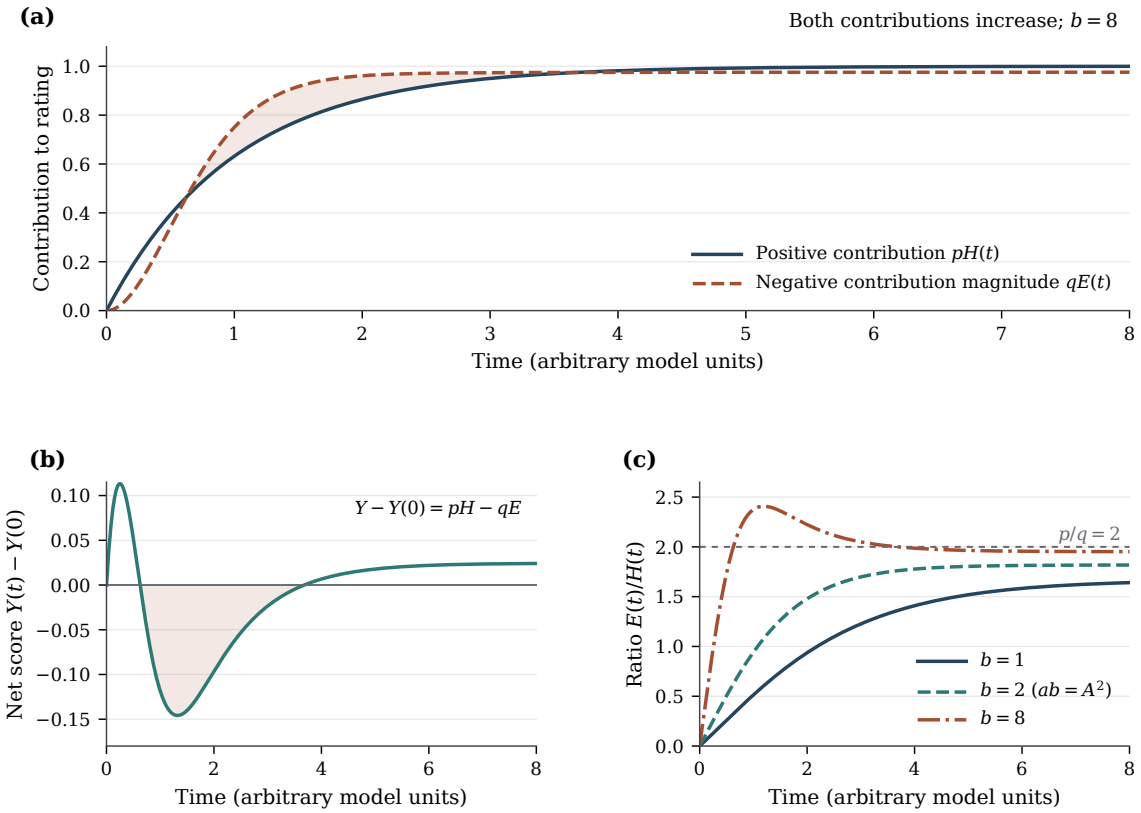


Figure 1: Synthetic recovery through a changing balance. With $a = d = 0.5$, $c = 0.1$, $p = 2$, $q = 1$ and $o = 0$, panel (a) follows the two increasing contributions pH and qE at $b = 8$, and panel (b) shows their difference. Panel (c) compares E/H for $b = 1, 2, 8$; the level $p/q = 2$ turns relative dominance into a baseline sign. The boundary case is $b = 2$. All curves start at $H = E = 0$; time is in arbitrary model units.

despite continuing to increase in absolute magnitude. The inequality is equivalently $bh > A$: the recruitment scale at familiarity's plateau exceeds familiarity's own relaxation rate. Saturation permits this reversal, but the rates decide whether it occurs.

The proof follows the same competition. General rules for monotonicity of ratios are established in Pinelis [5]; the particular coefficient and endpoint calculation below determine the boundary for this model. Differentiation gives

$$z' = b - D(H)z, \qquad D(H) = bH + c + \frac{a}{H} - A > 0. \tag{11}$$

Since $a/H - A = H'/H$, the coefficient D combines the opponent's adjustment coefficient $bH + c$ with the relative growth rate of familiarity. Growth of the denominator can therefore restrain E/H even while E itself increases. The ratio tracks $w = b/D(H)$. Initially $z = bt/2 + O(t^2)$ lies below $w = bt + O(t^2)$. Since $D_H = b - a/H^2$, the target rises for all finite times when $ab \le A^2$, keeping z below it and increasing. Above the boundary, the target turns down once. An asymptotic calculation shows that z eventually overtakes it, and an integrating-factor argument allows exactly one such crossing. Appendix B establishes these statements, including the equality case and the remainder bound for the asymptotics.

Above the boundary, a full recovery occurs exactly when

$$\frac{b}{K} < \frac{p}{q} < M, \qquad M = \max_{t>0} \frac{E(t)}{H(t)}. \tag{12}$$

The first inequality makes the final balance positive; the second allows the opponent to dominate temporarily. These conditions separate two questions that a single curve can obscure: whether

the dynamics permit a reversal of relative dominance, and whether the rating weights make that reversal cross the observed baseline. The removal coefficient c changes the limiting ratio and its peak, although it does not change the boundary between the two ratio shapes. Table 1 includes the other observation levels and the tangential case. The ratio result classifies baseline signs; it is not a classification of every local maximum or minimum of Y.

Table 1: Neutral-start score relative to baseline when $ab > A^2$. Here $L = p/q$, $z_\infty = b/K$ and $M = \max_{t>0} E(t)/H(t)$. The zero at $t = 0$ is excluded.

Observation level	Positive-time zeros	Negative values
$0 < L \le z_\infty$	One downward crossing	Yes
$z_\infty < L < M$	Downward then upward crossing	Yes
$L = M$	One tangential zero	No
$L > M$	No zero	No

Below or on the boundary, a negative value occurs exactly when $qb > pK$, with at most one downward crossing. At equality $qb = pK$, the score stays positive at every finite time and tends to zero. Thus $ab \le A^2$ is necessary and sufficient for this at-most-one-downward-crossing property and its negativity threshold to hold for every positive pair of observation weights. The shared-rate submodel has $d = b$, so $ab \le (a + b)^2/4$ places it wholly inside that region. Releasing the equality between loss and generation therefore changes a qualitative response possibility, not simply the freedom to fit a curve more closely.

For the example in equation (9), negativity at an intermediate time can also be proved without numerical integration. Let E_0 solve the same equation with $c = 0$ and the same H. Then

$$E_0(t) = 1 - \exp[-4(t - 1 + e^{-t})].$$

For $\delta = E_0 - E$, variation of constants gives $0 \le \delta(t) \le t/10$, because $\delta' = -8H\delta + E/10$ and $0 \le E \le 1$. It follows that

$$f(1) \le \tfrac{1}{10} - e^{-1} + e^{-4/e} < -0.038. \tag{13}$$

The positive initial slope, negative intermediate value and positive limit establish the recovered dip analytically. Proposition 1 then shows that these are the only two sign crossings.

4 What the recovered response can distinguish

The example offers a possible explanation of recovery: sequential recruitment followed by a reversal of relative dominance. To learn anything from that explanation, we must ask how much of it is needed. Would two ordinary linear relaxations already produce the same positive–negative–positive response, without the saturation in equation (2)? This comparison tests whether the observed pattern can distinguish the proposed interaction from a simpler specified account.

A constant-drive version of the sequential subsystem in Chou and D'Orsogna [7] provides a concrete comparison after reversing the sign of its negative component:

$$\dot{H} = a - AH, \qquad \dot{E} = bH - cE, \qquad f = pH - qE, \qquad H(0) = E(0) = 0. \tag{14}$$

Their published model uses a decaying drive and changes between intakes, so holding these fixed is an explicit protocol adaptation. Independent first-order relaxations belong to the same mathematical class considered here. More generally, a baseline-subtracted affine observation of a

stable two-dimensional affine system with constant input and real negative eigenvalues has one of the forms

$$f_L(t) = C + De^{-\lambda t} + Ge^{-\mu t}, \quad \lambda \neq \mu, \tag{15}$$

$$f_L(t) = C + (D + Gt)e^{-\lambda t}, \tag{16}$$

with positive decay rates and $f_L(0) = 0$. The coefficients may have either sign. The second expression includes a repeated eigenvalue and a possible Jordan contribution.

Proposition 2 (A restriction for two real decay modes)**.** *A nonzero function of either form* (15) *or* (16)*, with $f_L(0) = 0$, has at most one positive-time zero, and that zero is simple. If $f_L'(0) > 0$, a downward crossing occurs exactly when $C < 0$. When $C \geq 0$, the function remains positive at every finite positive time.*

Proof. For distinct rates,

$$e^{\lambda t} f_L'(t) = -\lambda D - \mu G e^{-(\mu-\lambda)t};$$

for repeated rates it is $G - \lambda D - \lambda G t$. Each has at most one zero unless $f_L' \equiv 0$, which would make $f_L \equiv 0$. Two positive-time zeros of f_L, in addition to its zero at the origin, would contradict Rolle's theorem. A nonsimple positive-time zero would also force two zeros of its derivative. A negative limit therefore forces one downward crossing after a positive initial slope, while a positive limit excludes any crossing. If $C = 0$, the forms reduce to $D(e^{-\lambda t} - e^{-\mu t})$ and $Gte^{-\lambda t}$, respectively; each has the fixed positive sign required by the initial slope. □

This elementary zero-count argument gives a limited comparator. The linear response has too few changes of direction to leave its initial baseline, cross it downwards, and then cross it upwards. The saturating cascade can do this with the same number of states, even though both states are themselves increasing. Thus its recovered dip supplies a qualitative distinction from this two-real-mode comparison class. It does not identify saturation as the only possible explanation: three real modes already suffice, because

$$g(t) = (1 - e^{-t})(e^{-t} - \tfrac{1}{4})(e^{-t} - \tfrac{3}{4}) \tag{17}$$

has $g'(0) = 3/16 > 0$, $g(\log 2) = -1/32 < 0$ and $g(\infty) = 3/16 > 0$. Expanding it gives a constant and three decaying exponentials.

A response shape can therefore exclude a specified account without uniquely selecting the proposed one. Delays, complex modes, changing inputs, nonlinear observations and heterogeneous mixtures fall outside the comparison; so do the broader history-dependent processes of Solomon and Corbit [3]. For a finite record, even the eventual limiting sign may remain unknown. Within those boundaries, the comparison identifies a response that the nonlinear two-state interaction permits. To interpret its components quantitatively, we must next distinguish what the output determines from what remains dependent on the chosen latent scale.

5 What component observations can determine

A valid observation of the opponent could directly show whether it increased or declined during recovery. That qualitative question is different from estimating every coefficient in the equations. The latter depends on which latent scale the observations fix, whereas the recovery prediction can be expressed through combinations that do not require that scale.

To see this, put $P = ab$, $u = bH$ and $\alpha = p/b$, retaining $A = a + d$. The same model becomes

$$u' = P - Au, \qquad E' = u(1 - E) - cE, \qquad Y = o + \alpha u - qE, \qquad u(0) = E(0) = 0. \tag{18}$$

Only the six combinations (A, P, c, α, q, o) determine this fixed-protocol response and its weighted contributions $pH = \alpha u$ and qE. For any $s \in (0, A)$, the original coefficients

$$a = s, \quad d = A - s, \quad b = P/s, \quad p = \alpha P/s \tag{19}$$

give those same trajectories, with c, q, o fixed. The positive state becomes $H = (s/A)(1 - e^{-At})$, so its magnitude varies while both contributions to the rating remain unchanged. This family has positive coefficients and keeps both states in the unit square.

The ambiguity does not prevent the present prediction of recovery. Exact observations of E and Y on any interval $[0, T]$, $T > 0$, identify all six reduced combinations under the model's neutral-start and known-time assumptions. Appendix C gives an explicit reconstruction. It follows that these traces determine both weighted contributions and the continuation of the response under the same fixed rules, even though they do not select the individual a, d, b, p in equation (19).

An independently calibrated trace of H, observed together with Y, additionally fixes the positive-state scale and identifies all seven original constants. Its proof is also given in Appendix C. An unknown but constant positive gain in an affine H proxy still permits recovery of the six reduced combinations after neutral-baseline subtraction. Thus absolute calibration is required for that full coefficient parameterization, not for the motivating fixed-protocol recovery prediction.

These statements concern uniqueness from exact trajectories. They provide neither a reliable high-derivative estimator nor a calibration of a psychological instrument. Near the start, the quantities used to separate recruitment and saturation appear at different small orders in time, and noisy finite records can be poorly informative. A study would need to justify both its observation map and its estimation method. There is no reason to require all seven coefficients when the scientific question concerns only the sign of the opponent's change or the response under the same protocol.

6 Initial state and behavioral interpretation

The neutral-start result supplies an explicit route to recovery. The positive response begins first, the recruited opponent may overtake it, and the positive contribution may later regain dominance while both states keep increasing. The ratio boundary identifies when the last reversal is possible. Its observation-weight interval then determines whether the reversal appears as a return above baseline, rather than a change of balance that leaves the net rating on one side of zero.

Other initial states produce different accounts of improvement within the same equations. If $H(0) = h$, familiarity remains fixed and

$$E(t) = E_* + [E(0) - E_*]e^{-Kt}. \tag{20}$$

An initial opponent above E_* then declines, so the rating improves through actual opponent reduction. Moreover, a neutral observed rating need not mean $H(0) = E(0) = 0$. For example, $a = 1$, $d = b = 2$, $c = 2$, $p = q = 1$ and $o = 0$, with $H(0) = E(0) = 3/4$, gives $Y(0) = 0$, $Y'(0) = -1/8$ and $Y(\infty) = 1/12$. Its initial decline and subsequent recovery occur even under the shared-rate restriction, because its latent start differs from the one used in Proposition 1.

The conditional behavioral inference is therefore specific. Recovery of the mixed rating is consistent with a growing opponent as well as with a declining one; the recovered curve alone does not choose between these accounts. Separate component observations already have a place in uncanny-valley research. For example, the repeated-interaction study of Złotowski and colleagues measured liking and eeriness separately and reported declining eeriness across encounters [8]. Such a result is evidence from its own measure, not an inference that the present model overturns.

Mapping that encounter protocol onto this continuous-time model would additionally require exposure durations, initialization and carryover assumptions.

The familiarity label also requires care. Recognizing an expected pattern, acquiring experience with a particular artificial object, and liking that object are different constructs. Here H is stipulated to contribute positively and recruit the opponent. Neither the equations nor the synthetic example establish that one measured construct has both roles. Likewise, an average across people need not obey the individual equations: the bilinear term involves $\mathbb{E}[HE]$, which generally differs from $\mathbb{E}[H]\mathbb{E}[E]$. The trajectory conclusions apply at the level where the model and its assumptions can be justified.

7 Conclusion

The two-state interaction permits a recovered negative excursion without any decline of its opponent. From a neutral latent start, this requires $ab > (a+d)^2$ together with $b/K < p/q < M$, where M is the unique maximum of E/H. Below or on the rate boundary, the opponent's relative contribution only increases. The shared-rate restriction excludes the recovered neutral-start crossing, and a constant-input system with only two real linear decay modes cannot reproduce it. These restrictions concern specified model classes and initial conditions; the three-mode example shows why the response does not identify a unique mechanism.

The model thus gives a conditional interpretation of recovery as a reversal of relative dominance. Its contribution is a worked mathematical account with explicit limits, rather than evidence that human eeriness follows these equations. Within the model, exact component information can determine the weighted response even while individual coefficients retain a scale ambiguity. The relevant observation should therefore be chosen for the question being asked. Establishing which account applies to people requires supported component meanings and compatible observations; a valley-shaped curve alone supplies neither.

AI assistance

ChatGPT (OpenAI, used during preparation in October 2026) assisted literature discovery and comparison, formulation and proof development, code generation, numerical checks, visualization and manuscript preparation. Separate AI-assisted checks examined mathematical arguments and readability. These checks are internal preparation and are not external peer review. The author remains responsible for the manuscript's claims, sources, calculations and final text.

Data and calculation statement

This theoretical study reports no new human or animal experiments, private participant data, or empirical parameter fits. Figure 1 consists of synthetic solutions of equations (1)–(2) with the stated parameters. The trajectories were computed with SciPy's DOP853 solver at relative tolerance 10^{-11} and absolute tolerance 10^{-13}. The sign bound in equation (13) establishes the example analytically, independently of that numerical illustration. All equations and parameter choices needed to reconstruct the displayed example are given in the paper.

A Convergence and monotonicity of the components

For arbitrary initial states in the square, $H - h = (H_0 - h)e^{-At}$. Put $e_* = B/K$ and $\eta = E - e_*$. Subtracting the equilibrium equation gives

$$\dot{\eta} = -(bH + c)\eta + b(1 - e_*)(H - h).$$

The coefficient $bH + c$ is at least $c > 0$. Variation of constants bounds $|\eta|$ by $|\eta(0)|e^{-ct}$ plus a constant times $\int_0^t e^{-c(t-s)}e^{-As}\,ds$. The integral tends to zero, including when $A = c$, where it equals te^{-ct}. Thus every trajectory converges to the stated equilibrium.

For a neutral start, the target $T = bH/(bH + c)$ has $T' > 0$. The difference $v = T - E$ satisfies $v' = T' - (bH + c)v$ and $v(0) = 0$. Its integral solution is positive for $t > 0$, so $E' = (bH + c)v > 0$. This proves component monotonicity without asserting monotonicity of their weighted difference.

B Proof of the ratio classification

One can connect the proof directly to the ratio rules in Pinelis [5]. Put $I(t) = \int_0^t [bH(s) + c]ds$, $\mathcal{F} = Ee^I$ and $\mathcal{G} = He^I$. Then $\mathcal{G}' > 0$ for $t > 0$, $\mathcal{F}/\mathcal{G} = z$, and $\mathcal{F}'/\mathcal{G}' = b/D(H)$. The shape of this derivative ratio supplies the general monotonicity structure (Propositions 4.1–4.4 of that source). The following direct argument includes the model-specific endpoint calculation that selects the unique maximum above the boundary.

For $0 < H < h$, equation (11) has $D > 0$, because $a/H > A$. Set $w = b/D(H(t))$. Taylor expansion gives $H = at + O(t^2)$, $E = abt^2/2 + O(t^3)$, and hence $z = bt/2 + O(t^2) < w = bt + O(t^2)$ initially. If $ab \le A^2$, then

$$D_H = b - \frac{a}{H^2} < b - \frac{A^2}{a} \le 0$$

for every finite positive time. Thus $w' > 0$. At a first contact from below, $(z - w)' = -w' < 0$, which is impossible. It follows that $z < w$ and $z' > 0$ throughout. Convergence gives $z \to b/K$. The strict inequality $H < h$ includes the boundary case $ab = A^2$.

If $ab > A^2$, the value $H_c = \sqrt{a/b}$ lies in $(0, h)$ and is reached once, at time t_c. The target w increases before t_c and decreases afterward. The same barrier argument, or the integral solution of $(z - w)' + D(z - w) = -w'$, gives $z(t_c) < w(t_c)$. To determine what happens later, put $x = e^{-At}$. Here $B > A$ and $K > A$. For $\eta = E - B/K$,

$$\eta' = -(K - Bx)\eta - \frac{Bc}{K}x.$$

Define $C_0 = -Bc/[K(K - A)]$ and $r = \eta - C_0 x$. Direct substitution yields

$$r' = -(K - Bx)r + BC_0x^2.$$

The propagator is bounded by $e^{B/A}e^{-K(t-s)}$, so

$$|r(t)| \le e^{B/A}\left[|r(0)|e^{-Kt} + |BC_0|\int_0^t e^{-K(t-s)}e^{-2As}\,ds\right].$$

The integral is $(e^{-2At} - e^{-Kt})/(K - 2A)$ when $K \ne 2A$ and te^{-2At} when $K = 2A$. Since $K > A$, the remainder is $o(e^{-At})$ in both cases. Therefore

$$z = \frac{b}{K}\left[1 + \frac{B - A}{K - A}x\right] + o(x), \tag{21}$$

$$w = \frac{b}{K}\left[1 + \frac{B - A}{K}x\right] + O(x^2). \tag{22}$$

The second expansion follows from $D = K + (A - B)x + Ax^2/(1 - x)$. Both $z - b/K$ and $z - w$ are positive for sufficiently large time.

On (t_c, ∞), let $g = z - w$. The equation $g' + Dg = -w' > 0$ makes g times its positive integrating factor strictly increasing. It starts negative and eventually becomes positive, so it has exactly one zero. Since $z' = -Dg$, the ratio increases and then decreases, with a unique maximum above its limit. At the zero, $g' = -w' > 0$ and $z'' = -Dg' < 0$, establishing nondegeneracy. Equation (10) now gives every case in Table 1. In particular, the tangential zero at $p/q = M$ is a minimum of the score with no sign change.

C Parameter information and prediction

Exact opponent and mixed-response traces

The reduced equations (18) are analytic. Write $e_j = E^{(j)}(0)$ and $y_j = Y^{(j)}(0)$. Repeated differentiation gives

$$\begin{aligned} e_2 &= P, \\ e_3 &= -P(A + c), \\ e_4 &= P(A^2 + Ac + c^2) - 3P^2, \\ e_5 &= -P(A^3 + A^2c + Ac^2 + c^3) + P^2(10A + 7c). \end{aligned}$$

Consequently the exact E trace determines

$$P = e_2, \qquad S = -e_3/P = A + c,$$

$$J = S^2 - e_4/P - 3P = Ac, \qquad A = \frac{e_5/P + S(S^2 - 2J) - 7PS}{3P}, \qquad c = S - A.$$

The denominator is nonzero since $P > 0$, including when $A = c$. With $u = P(1 - e^{-At})/A$, the output gives

$$o = Y(0), \qquad \alpha = y_1/P, \qquad q = (-y_2 - Ay_1)/P.$$

Thus all six reduced combinations are globally unique. Every original positive coefficient choice that gives the same E, Y must have the form (19), and each such choice does give the same traces. This establishes the full remaining ambiguity for these observations. Invariance of a combination alone would not have proved its identifiability.

For clarity, the recovery criterion also has a scale-free form. Set $r = E/u$, let $R = \max_{t>0} r(t)$ in the overshoot regime, and write $K = P/A + c$. Then $r(\infty) = 1/K$ and the recovered crossing requires

$$P > A^2, \qquad 1/K < \alpha/q < R.$$

These are exactly the original conditions expressed in identified combinations. A selective intervention on an individual rate would require a further input law; the fixed-protocol argument does not determine responses to arbitrary changes of exposure.

A calibrated positive component

Exact calibrated H and Y determine $a = H'(0)$, $A = [a - H'(t)]/H(t)$ for $t > 0$, and $d = A - a$. Neutrality gives $o = Y(0)$ and $p = Y'(0)/a$. The known residual $W = o + pH - Y = qE$ obeys

$$W' = (qb)H - bHW - cW. \tag{23}$$

Identifying the constant coefficients of this linear-in-parameters equation is the setting of the feature-span criterion in Stanhope, Rubin and Swigon [6, Theorem 5.3]. Here the required

independence follows directly. If $r_0H + s_0HW + v_0W = 0$, division by H and passage to $t \downarrow 0$ gives $r_0 = 0$, since $H \sim at$ and $W = O(t^2)$. Dividing the remainder by $W > 0$ gives $s_0H + v_0 = 0$; strict increase of H forces $s_0 = v_0 = 0$. The trace therefore determines qb, b, c, and then q, proving uniqueness of all seven original constants.

If only an affine proxy $Z = z_0 + r_H H$ with unknown constant $r_H > 0$ is observed alongside Y, the neutral start still fixes $z_0 = Z(0)$. Put $X = Z - Z(0)$, $F = X'(0)$ and $v = Y'(0)/F$. The known residual $W = o + vX - Y = qE$ satisfies

$$W' = \theta X - \beta XW - cW, \qquad \beta = b/r_H, \quad \theta = q\beta.$$

The same independence argument determines θ, β, c. Then $A = [F - X'(t)]/X(t)$ for $t > 0$, $P = F\beta$, $\alpha = v/\beta$, and $q = \theta/\beta$. The reduced prediction is therefore identified without the absolute proxy gain. Neither calculation supplies a practical estimator for noisy data or justifies a nonlinear questionnaire-to-state mapping.

D Static curve freedom and prior state architecture

If the rate functions of appearance are unrestricted, any smooth equal-weight equilibrium affinity $f(R) \in (-1, 1)$ can be encoded by positive rates. For an arbitrary inverse-time scale $\rho > 0$, set

$$a = \rho(1 + f), \quad d = b = \rho(1 - f), \quad c = \frac{\rho}{2}(1 + f)^2. \tag{24}$$

Then $H_* = (1+f)/2$ and $E_* = (1-f)/2$, so $H_* - E_* = f$. This construction concerns unrestricted functions, not every prespecified finite-dimensional rate family. It explains why reproducing a static curve without constraints on the rates cannot identify its underlying process. Equilibrium components fix $d/a = (1 - h)/h$ and $c/b = h(1 - E_*)/E_*$, but not a common time scale, since multiplying all four rates by a positive constant preserves the equilibrium.

For fixed input, put $X = bH + c$ and $Q = 1 - E$. The state equations become

$$\dot{X} = P + Ac - AX, \qquad \dot{Q} = c - XQ, \qquad P = ab. \tag{25}$$

This is a constant-input member of the bilinear feedforward family in Sontag [4]. The transformed observation is $Y = o - q - (p/b)c + (p/b)X + qQ$. The correspondence credits the prior state dynamics while identifying the mixed observable analyzed here; it supplies no biological identification of the states.

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